NETWORKING

HTTP/2 & HTTP/3 Visual Guide

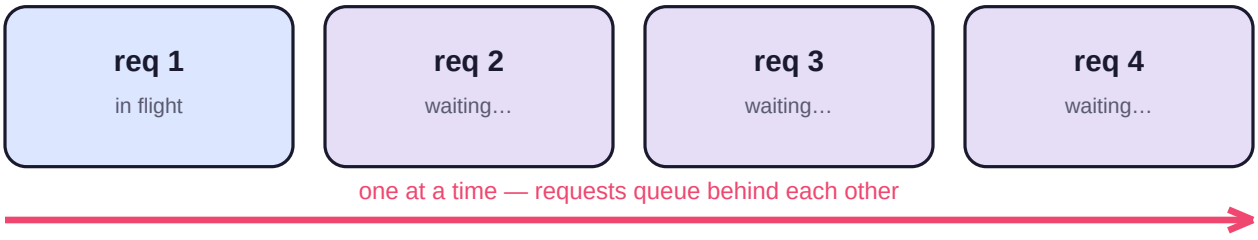
Beyond HTTP/1.1 —
multiplexing, header compression, and QUIC.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

HTTP/1.1: Head-of-Line Blocking

One request at a time per connection — the browser opens 6+ to cope.

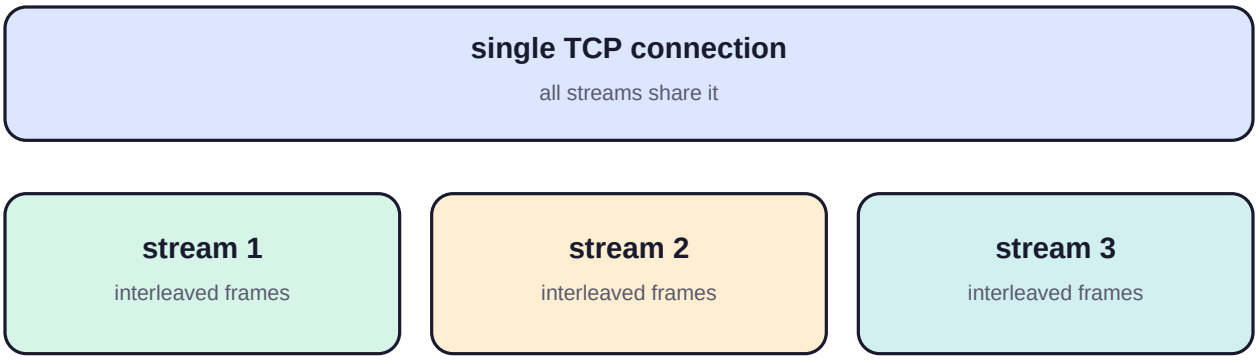


- Text protocol, one request per connection at a time — a slow asset blocks everything behind it.
- Workarounds: domain sharding, spriting, inlining — hacks to dodge the protocol's limits.
- Keep-alive reuses connections but doesn't fix the queueing. Pipelining was broken in practice.

STEP 2/5

HTTP/2: Multiplexing

Many streams, one connection — binary framing kills the queue.



- Binary framing: requests split into frames, multiplexed by stream ID — no head-of-line blocking between streams.
- Server push (mostly abandoned) and stream priorities: the browser hints what matters first.
- One catch: TCP-level head-of-line blocking remains — one lost TCP packet stalls ALL streams. That's what HTTP/3 fixes.

STEP 3/5

HPACK & QPACK: Header Compression

HTTP headers are repetitive — stop resending them.

HPACK (HTTP/2)

dynamic table: 'user-agent: …' sent once, referenced by index after

QPACK (HTTP/3)

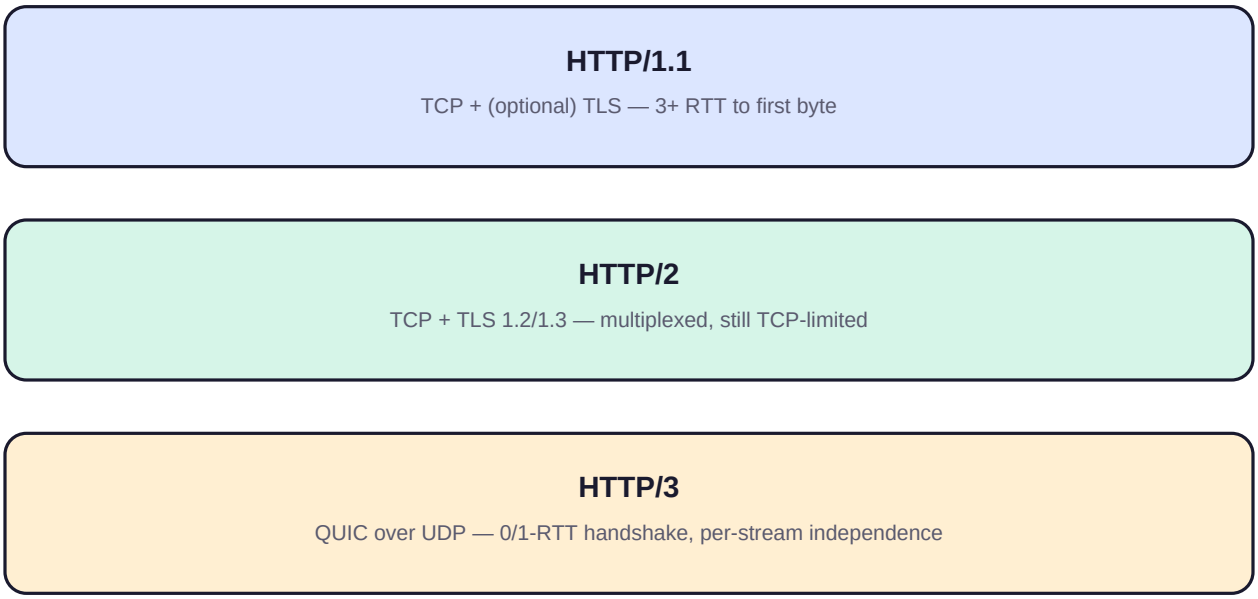
same idea, redesigned for QUIC's out-of-order streams

- Typical savings: hundreds of bytes per request → a few. Huge on mobile and high-request pages.
- Static table covers common headers; the dynamic table learns per-connection patterns.
- CRIME-style attacks taught a lesson: compression + attacker-controlled input needs careful design.

STEP 4/5

HTTP/3: QUIC Transport

Drops TCP entirely — runs over UDP with TLS 1.3 baked in.

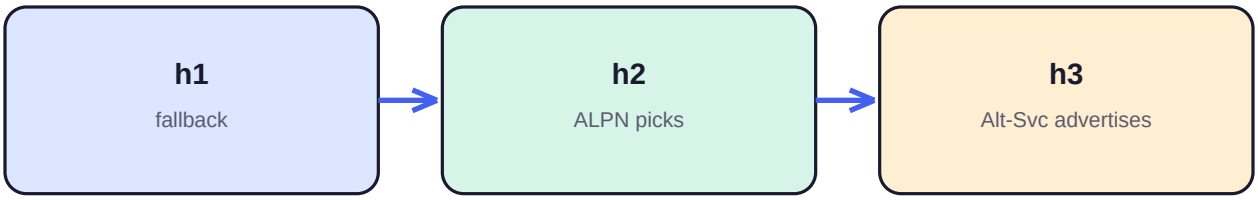


A lost packet stalls only ITS stream now — TCP's shared fate is gone. Connection migration survives IP changes (W

STEP 5/5

The Upgrade Path

Negotiation is automatic — servers and browsers agree on the best version.



- TLS ALPN negotiates h2 during the handshake; Alt-Svc headers advertise h3/QUIC availability.
- Deployment: enable h2/h3 on your CDN or load balancer first — origin can lag behind safely.
- Measure: multiplexing helps most on high-latency, lossy, or asset-heavy pages.

STEP ✓/✓

Key Takeaways

HTTP/2 & HTTP/3 — Remember

- HTTP/1.1 serializes requests per connection — head-of-line blocking forced hacks like sharding.
- HTTP/2 multiplexes binary-framed streams over one TCP connection.
- HPACK/QPACK compress repetitive headers to a few bytes via indexed tables.
- HTTP/3 replaces TCP with QUIC over UDP: 0/1-RTT, per-stream loss independence, connection migration.
- Negotiation is automatic (ALPN, Alt-Svc) — deploy at the edge first.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev